

Project Design Phase-II

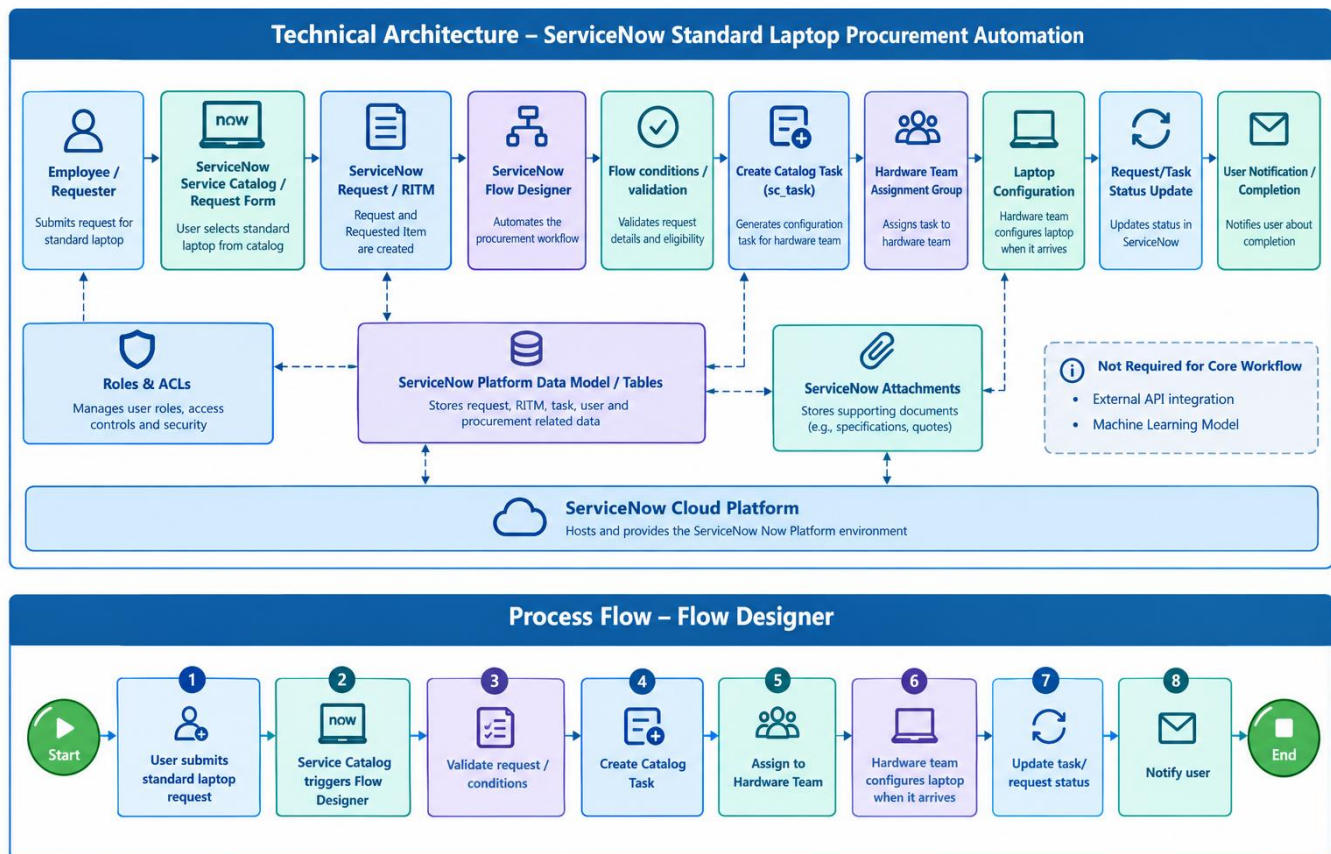
Technology Stack (Architecture & Stack)

Date	30 September 2026
Team ID	SWTID-2026-9263
Project Name	Streamlining IT Procurement: Automating Standard Laptop Orders with Flow Designer
Maximum Marks	4 Marks

Technical Architecture:

The proposed solution is implemented on the ServiceNow Platform. Employees submit a standard laptop request through the ServiceNow Service Catalog. Flow Designer is triggered by the catalog request and applies the required conditions and validation before creating a Catalog Task (sc_task). The task is assigned to the Hardware Team assignment group for laptop configuration. After configuration, the task/request status is updated and the user is notified. ServiceNow platform tables/data, attachments, roles and access controls support the workflow.

Architectural Diagram – ServiceNow Standard Laptop Procurement Automation



Core flow: Employee → Service Catalog → Request/RITM → Flow Designer → Validation → Catalog Task → Hardware Team → Laptop Configuration → Status Update → User Notification.

Table-1 : Components & Technologies:

S.No	Component	Description	Technology
1	User Interface	Interface through which employees submit the standard laptop request and enter the required request information.	ServiceNow Service Catalog / Service Portal
2	Application Logic-1	Receives the submitted catalog request and starts the automated fulfillment process.	ServiceNow Flow Designer
3	Application Logic-2	Applies conditions/validation and creates the hardware configuration task required to fulfill the request.	Flow Designer Conditions + Actions
4	Application Logic-3	Assigns the generated catalog task to the appropriate hardware team and controls task/status progression.	Flow Designer + ServiceNow Task Management / Assignment Groups
5	Database	Stores users, requests, requested items, catalog tasks, assignments and workflow-related records.	ServiceNow Platform Data Model / ServiceNow Tables (e.g., Request, Requested Item and Task records)
6	Cloud Database	Provides the cloud-hosted data persistence used by the ServiceNow instance; direct database administration is not required by the application.	ServiceNow Now Platform Cloud Data Layer
7	File Storage	Stores supporting attachments such as laptop specifications, quotations or other procurement documents when required.	ServiceNow Attachment Storage
8	External API-1	No external API is required for the core standard laptop procurement workflow.	Not Applicable
9	External API-2	No second external API is required for the core workflow. Future integrations can be added if required.	Not Applicable / Optional REST Integration
10	Machine Learning Model	The proposed workflow is rule-based and does not require a machine learning model.	Not Applicable
11	Infrastructure (Server / Cloud)	The workflow is hosted and executed within the ServiceNow cloud environment; no separate application server is required for the core solution.	ServiceNow Now Platform – Cloud

Table-2: Application Characteristics:

S.No	Characteristics	Description	Technology
1	Open-Source Frameworks	No separate open-source application framework is required for the core solution. The workflow is configured using ServiceNow platform capabilities and Flow Designer.	ServiceNow Flow Designer / ServiceNow Now Platform

S.No	Characteristics	Description	Technology
2	Security	Access to requests, tasks, attachments and application data is controlled using roles and Access Control Lists (ACLs). HTTPS/TLS protects data in transit, while ServiceNow provides platform-level encryption and security controls.	ServiceNow RBAC, ACLs, HTTPS/TLS, Platform Encryption
3	Scalable Architecture	The solution uses a cloud platform and reusable workflow automation. The same Flow Designer process can handle multiple standard laptop requests without creating a separate application instance for each request.	ServiceNow Now Platform + Flow Designer
4	Availability	The application runs on the ServiceNow cloud platform rather than on a project-managed local server. ServiceNow provides cloud-hosted platform infrastructure and high-availability capabilities.	ServiceNow Now Platform Cloud
5	Performance	Automation reduces manual hand-offs by creating and assigning the hardware task immediately after the qualifying request is submitted. This reduces processing delay and improves fulfilment throughput.	Flow Designer Automation + ServiceNow Task Management

References:

ServiceNow – Now Platform Reference Guide: <https://www.servicenow.com/content/dam/servicenow-assets/public/en-us/doc-type/resource-center/ebook/ebk-now-platform-reference-guide.pdf>

ServiceNow – Securing the Now Platform: <https://www.servicenow.com/content/dam/servicenow-assets/public/en-us/doc-type/resource-center/ebook/ebk-how-servicenow-delivers-safe-and-secure-cloud-services.pdf>

ServiceNow Developers – Now Platform / Data Model: <https://developer.servicenow.com/>

ServiceNow – C4 Model reference used by the template: <https://c4model.com/>

IBM Cloud Architecture reference retained from the template: <https://www.ibm.com/cloud/architecture>

AWS Architecture reference retained from the template: <https://aws.amazon.com/architecture>